

D.E. Shaw Interview Experience

D.E. Shaw & Co., also known as the D. E. Shaw Group, is a prominent global investment and technology development firm. Founded in 1988 by David E. Shaw, a former Columbia University computer science professor, the firm has established itself as a leader in quantitative investing, proprietary trading, and technology-driven financial strategies. D.E Shaw India was founded in 1996 and it has offices at Hyderabad, Bengaluru and Gurugram.

Role Offered: SDET Intern

Stipend: 1,50,000 INR per month

Sequence of Selection: PPT, Online Assessment, Technical Interview, HR Interview

Online Assessment:

It consisted of three sections:

- Coding Question – 2 (20 mins + 30 mins)
- Technical MCQs – 12 (17 mins)
- General Aptitude – 14 (28 mins)

The test lasted for 95 mins and an important point to note is that you can't go back to previous section once you have submitted it, and even if you submit one of the sections earlier than the stipulated time, the remaining time will not be carried forward.

Coding Question 1:

It was a slight variation of [Print all possible words from phone digits](#) problem. Instead of printing all possible words, there will be a set of words and a sequence of digits, we need to find the number of words out of the given set that can be formed by pressing the given sequence of digits.

Sample Testcase:

contacts = ["ehcv", "fibo", "dgat"]

pattern = 3428

Output: 2

Solution:

- 3 – 'd', 'e', 'f'
- 4 – 'g', 'h', 'i'
- 2 – 'a', 'b', 'c'
- 8 – 't', 'u', 'v'

Only two contacts ("ehcv" and "dgat") can be formed by pressing 3428 buttons. So, the answer is 2.

Coding Question 2:

Given an array of weights and profits. You can remove as many weights as possible to make the weights array as non-decreasing. You need to make the weights array non-decreasing such that the profit is maximum.

Sample Testcase:

weights = [1,3,2,4,2]

profit = [5,6,8,7,9]

Output: 22

Explanation: Removing weights[1] and weights[3] makes the weights array non-decreasing and maximizes the profit.

Technical MCQs:

Technical MCQs were easy to medium level, and the questions were based on predicting output, identifying the line where logical error occurs, DSA and DBMS.

General Aptitude:

I couldn't remember many of the questions but there was a logical reasoning kind of question with 1 statement and 4 questions based on that, which was little bit similar to [Logical Reasoning](#) (but more complex than this) and was difficult. Except those 4 questions, others were easy to medium level.

Marking Scheme for MCQs: +3/-1

Based on this online assessment, 21 people were shortlisted for the next round.

Technical Interview – 1: (1 hour)

The interviews were conducted at CUIC. The interview started off, and I was asked about my strong subject. I told them I am equally strong in both DBMS and OS. Then the interviewer asked about the advantages of DBMS over file system and he said that even DBMS stores data in files then how it is advantageous? I was not able to answer the way the interviewer wanted to be. Then they started to check my problem-solving skills.

1. Given an array of integers, find the number of triplets with even product.
Initially, I gave a wrong answer but the interviewer corrected me and I was able to give the correct solution.
2. Given an array of integers, where each index represents an animal and arr[index] represents its predator. Find the minimum number of cages required so that none of the animals will be killed by others. (Transitive relation is also possible (i.e) if 2 is predator of 3 and 1 is predator of 2, then 1 is also a predator of 3)

I told them that we can solve using graphs but the interviewer wanted me solve in $O(1)$ space complexity and he asked me what kinds of input might be edge cases that we need to consider and I was asked to write the code on paper.

3. Given two strings s_1 and s_2 , find minimum length substring in s_1 which contains all characters of s_2 in it.

I was able to give the solution and write the code, but the interviewer asked me to reduce its time complexity but I was not able to do that.

Then the interviewer checked my OS knowledge. They asked, In IDEs like VSCode, how code editor and terminals run at the same time and how terminal executes background processes and all. I was not sure about the answer and I explained to them about processes, threads and all but I was not able to give the accurate answer.

Then the interviewer asked me to convert given postfix to prefix directly but I couldn't answer that. Then the interviewer asked about OOPS like virtual function, abstract methods, run-time polymorphism and all and asked me to write code related to that.

Then, finally I was given a question related to binary search. Given an 2D matrix sorted in spiral order, write algorithm to find an element. I gave them the brute force $O(n^2)$ solution but the interviewer asked me to optimize it and asked me to tell what came to mind, and I told them we could use binary search but I was not able to tell the correct logic.

Finally, the interviewer asked whether I had any questions for them, and I asked about the challenges that an intern might face during their internship.

There wasn't any specific shortlisting, and the students who performed well were asked to attend another round of interviews. I was the last one to be called for another round of interview at 8:15pm.

Technical Interview – 2: (45 mins)

The interview started off with self-introduction (in my first round of interview they haven't even asked to introduce myself). Since I have mentioned that I am good at React and Node in my resume, the interviewer asked the following questions:

The interviewer first asked why Node was preferred by most of the programmers, even though there are a lot of options available for the backend. Then he asked that there is a web application where Node from backend pushes data each 1 sec and front end is developed using React whose state is maintained by Redux. Is it possible to check whether node pushes data every one second? If so, how? I was not sure about this question, so we moved on.

Then the interview asked some basic questions about process in OS and how process data are maintained. Then they asked about various scheduling algorithms and how processes undergo execution. Then they asked which scheduling algorithm you would use to maximize resource utilization.

Then they started off with DSA questions.

1. Given an array in which all elements except one occur an even number of times.
I gave them the efficient XOR approach. XOR of same numbers is 0 so all the numbers which occur even number of times get cancelled out by taking XOR for all numbers. Then immediately I was given another question related to XOR.
2. Given an array of elements and integer k, we need to remove k elements to maximize the XOR of all elements.
I was only able to give the brute force approach to this question.
3. Given an array of integers and an integer k, find all triplets with sum less than or equal to k.
I first told them the brute force method and, then gave them the three-pointer approach.
4. Given an array of pages and intelligence and an integer k, where pages represent no of pages in each book and intelligence represent amount of knowledge you gain by reading that book, k represents maximum pages you can read in a day, find the maximum intelligence you can gain in a day.
It was similar to 0/1 knapsack problem. I was asked to write the code on paper once I gave them the solution.
5. Given a 2D grid of characters and a word, return true if the word exists in the grid. From one cell you can move to any of the four neighbouring cells.
I gave them backtracking solution and wrote the code and the interviewer was happy with that.

Then the interviewer wrote some code and asked me to find the time complexity, guided me to the correct answer and asked about the order in which constructor and destructor get executed in inherited sub classes. With that the interview got over.

I had no HR interview.

The result was announced the next day, and I was the only one who got the offer.

General Tips:

- Regular practice is important. Solve problems in Leetcode, GeeksForGeeks and Hackerrank. A lot of people don't prefer Hackerrank, but at least get familiar with it because the online assessment will be conducted on Hackerrank.
- Follow C.O.D.E resources. It will help you to get an idea about various types of problems in each topic. And I followed takeUForward YouTube channel for the topics like Binary Search, Graphs, Trees and DP.
- There is no common method or strategy to prepare for interviews. You need to find your own strategy which makes you feel confident in solving problems.

- If you are not confident with DBMS, OS and OOPS, then prepare for them simultaneously because you won't be having much time between the online test and the interview.
- Aptitude is as important as DSA. At least start learning aptitude two weeks before your online test.
- Even though I was not asked about my projects, in every interview they will ask about your projects, and they might ask you about the database structure and all. So be prepared to answer any questions related to your projects.
- A day before your online test, refer to previous year questions asked by the company that might get repeated.
- Explain to the interviewer about the logic and how you came up with this solution. Most of them are interested in your thought process, not the code.
- Initially, for any DSA question, first give the brute force approach and then tell how you can optimize that.
- Whenever you learn a new concept and if you feel you might forget it, please make a note of it (either handwritten or a word doc) and maintain it throughout your preparation which will help you during your last-minute revision.

All the very best.

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